VF Asia Footwear Laboratory Testing Manual

Test Method for Edgewise Crush Strength	Test method # VF- B27 Page 1 of 3
Date Issued: May,2019 Revision #: 1 Reason for revision: Rectify the unit of measure from kg to kgf for strength of material	Created by: Nila Babia Erazo

PURPOSE AND SCOPE: To measure the ability of corrugated cardboard / outer-box to resist crushing under load applied vertically to an edge with the flutes perpendicular between two flat platens.

REFERENCES: SATRA TM440

EQUIPMENT AND MATERIAL:

1. Tensile testing (compression) machine, meeting the requirements of ASTM D638
2. Compression platens, one of which can move towards the other to compress a test sample between them.
3. Sharp cutter, straight and parallel-sided perpendicular to the board surfaces.
4. A template for checking the dimension, square-ness and edge perpendicularity of the test pieces
5. Two rectangular smooth-finished metal guide blocks: 100±5 mm long and 20±1 mm square cross -section with device to enable them to be pulled clear of the test specimen during the test.
6. Corrugated cardboard, paperboard, outer-box, box

SAMPLE PREPARATION:

1. Condition the sample for 24 hours at standard laboratory conditions @ 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.
2. Cut the samples with dimensions of 5.0 ± 0.1 cm square.
   - The test piece shall be cut from undamaged areas and shall not include any existing edges.
   - The test piece shall be cut away from folds or joins and not contaminated with adhesives or closure tapes.
3. Inspect the edges of test specimens. Reject any that are not free from loose fibers on their edges or have one or more sides with non-parallel edges.
4. After cutting, the test piece edges should NOT be handled and must ensure the edges are not in contact with any other objects. No significant pressure should be placed on the specimens in any directions.

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5. Do not mark or label the test pieces before testing – this will affect their surfaces which could affect their performance. Marking will be made after testing.
6. Check the dimensions of the faces of each test piece by placing each in turn on the template with dimension 5.0mm x 5.0 mm.
7. Condition the test pieces for at least 3 hours at standard laboratory conditions.

REPETITIONS: Test three (3) samples per material.

TEST PROCEDURE:

Test should be conducted at standard laboratory conditions @ 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

1. Adjust the compression / tensile testing machine so that the platens are approximately 70 mm apart.
2. Position the test plate vertically at the center of the bottom platen such that the axes of the flutes are perpendicular to the surface of the platens, meaning the waves of the flute sections are visible at the top of the test piece.
3. Place the two guide blocks, either side of the test piece to support the test piece in vertical alignment. These should be placed so that they are just touching the test piece without exerting any pressure on it.
4. Tare the compression / tensile testing machine. Set the crosshead speed to 10 mm/min
5. When the load is between 4.0 – 5.0 kgf, carefully move the guide blocks away from test piece by means of the devices attached to their sides. Do not stop the compression test machine and do not remove the guide blocks from the bottom platen.
6. Terminate the test after buckling has occurred and a maximum load recorded.
7. Record the maximum load before failure to the nearest 0.1 kgf. Record also the mode of failure, for example, the position of the buckle line(s), any delamination of the corrugated layers.

   Note: If the test piece slips forward or backwards and show no buckling, the test shall be considered invalid and the test piece discarded.

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8. Mark the test piece with a unique identifier.
9. Repeat the steps for the remaining test specimens.
10. Record the maximum load to the nearest 0.1 kgf

CALCULATIONS:

Edgewise Crush Strength (kgf /m) = F/ W

F = load in kilogram.

W = width of test plate in meter